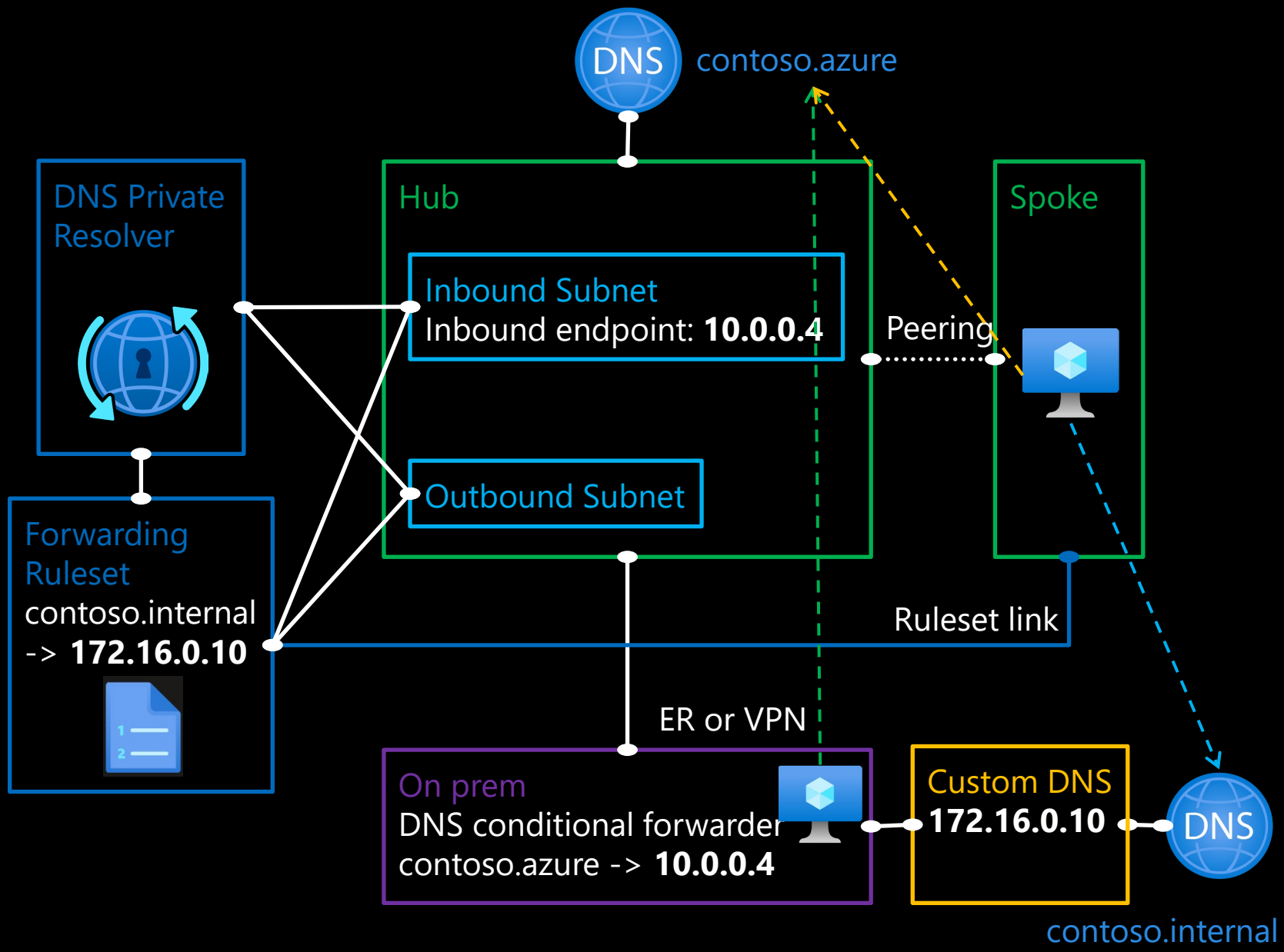
DNS Private Resolver



DNS resolution challenges

Customers need to deploy and maintain custom DNS servers on Azure VMs and set up forwarding rules for:

- On-prem to Azure resolution
- Azure to on-prem resolution
- Azure to Azure resolution

Maintenance of these VMs is a big customer pain-point.

Introduction to DNS Private Resolver

PaaS service.

Enables users to query Azure DNS Private zones from on-prem environment without deploying VM based DNS servers.

It could also query Custom DNS servers deployed on-prem or in Azure.

It acts as a custom forwarder, not as DNS Zone.

It forwards the DNS query to the server that is authoritative of the domain name.

Components of DNS Private Resolver

Injected into a VNET (1:1 relationship), typically the Hub.

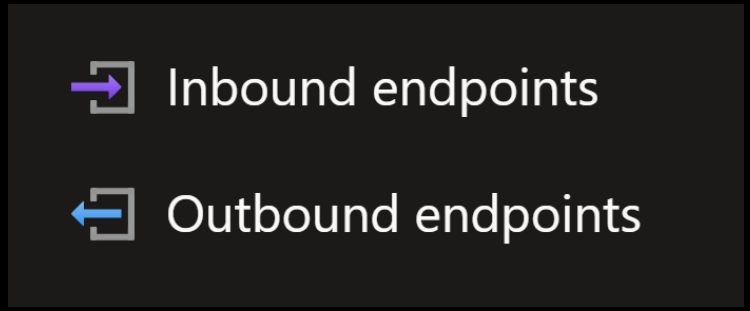
Need (at least) one dedicated subnet to provision inbound endpoint(s).

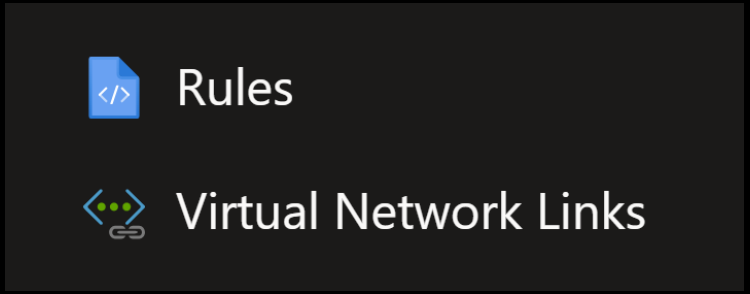
Need zero or more dedicated subnet to provision outbound endpoint(s).

Need zero or more DNS Forwarding Ruleset.

Group of DNS Forwarding Rules (up to 1,000), applied to one or more Outbound Endpoints, or linked to one or more VNet (1:N relationship).

Virtual Network link(s) to enable resolution from linked VNET.





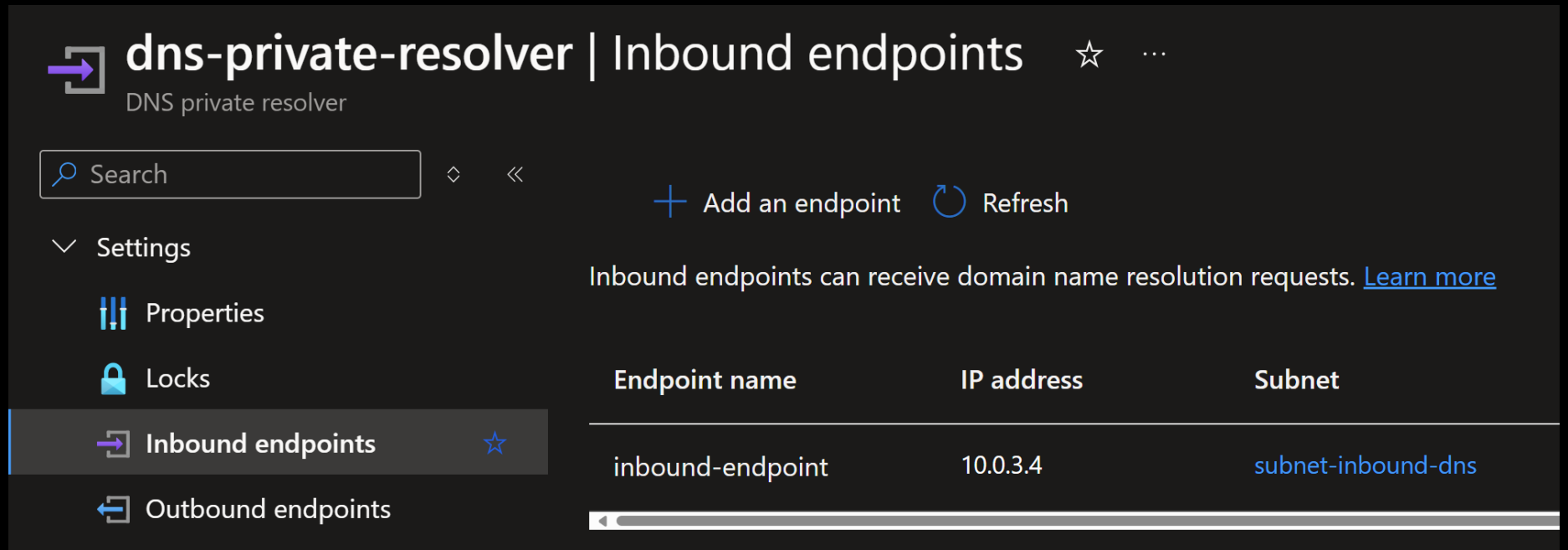
DNS Private Resolver - Inbound endpoints

Uses a dedicated Subnet with at least /28 and recommended /24.

Exposes a private IP address (VIP) that will be used to forward DNS queries.

Inbound Endpoint resolves all Private DNS Zones linked to the VNET where it is deployed.

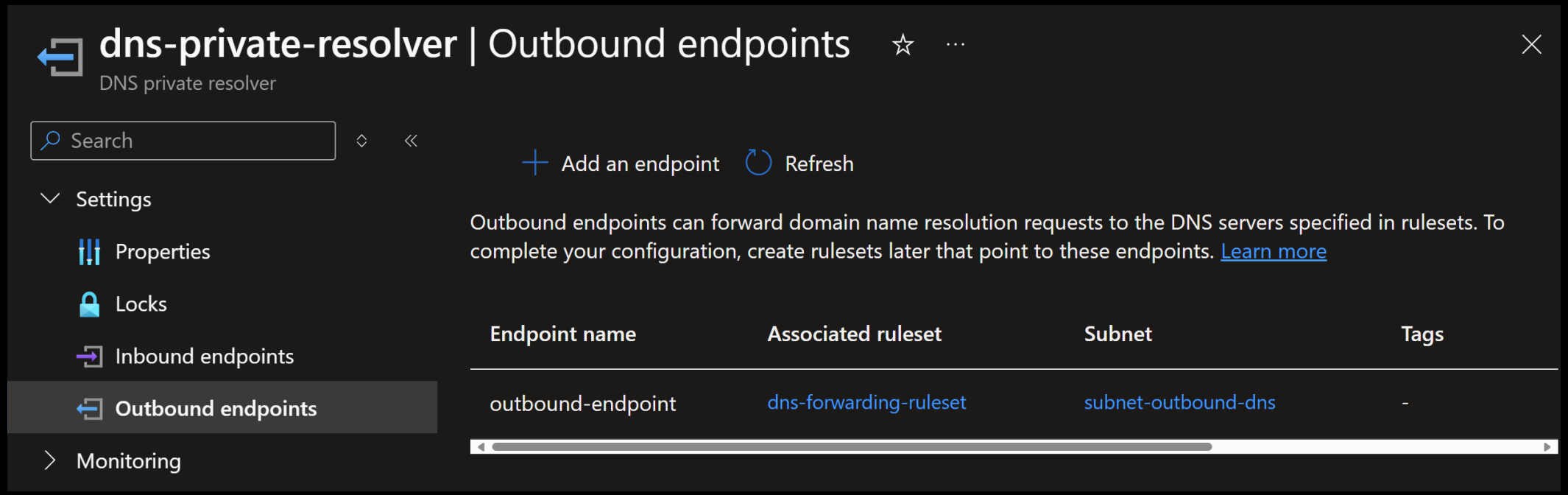
Used to resolve Private DNS Zones from on-prem to Azure.



DNS Private Resolver - Outbound endpoints

Enables conditional forwarding name resolution from Azure to on-premises, other cloud providers, or external DNS servers.

Uses a dedicated Subnet with at least /28 and recommended /24.

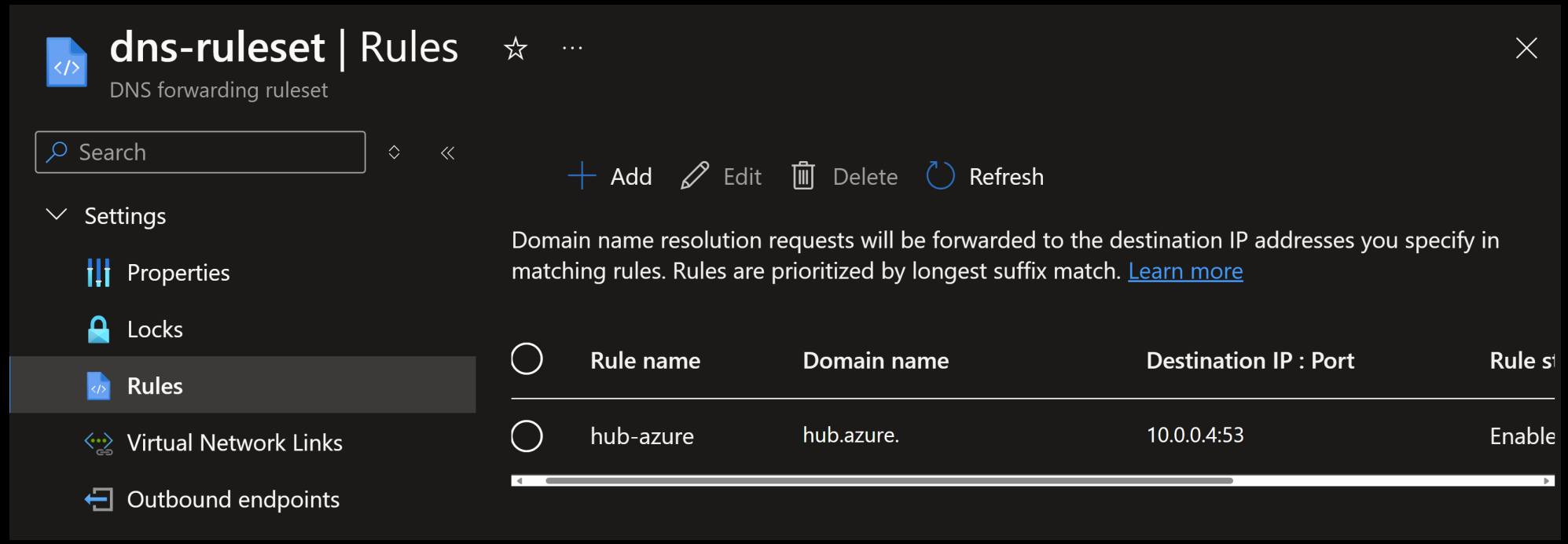


DNS Private Resolver - Forwarding rulesets

"If you want to resolve domain name *hub.azure*, go to *10.0.0.4:53*"

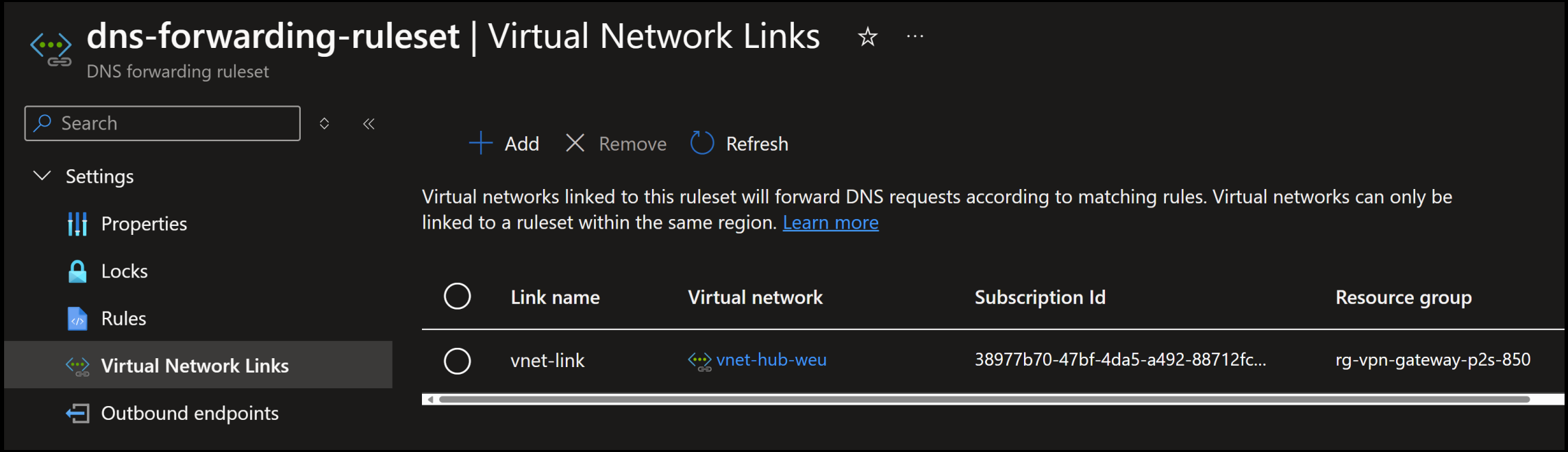
Group of DNS forwarding rules (up to 1000).

Is linked to one or more VNETs, enabling them to resolve the domain names.



DNS Private Resolver - Virtual network link

Enables name resolution for virtual networks that are linked to an outbound endpoint with a DNS forwarding ruleset.



DNS Private Resolver pricing

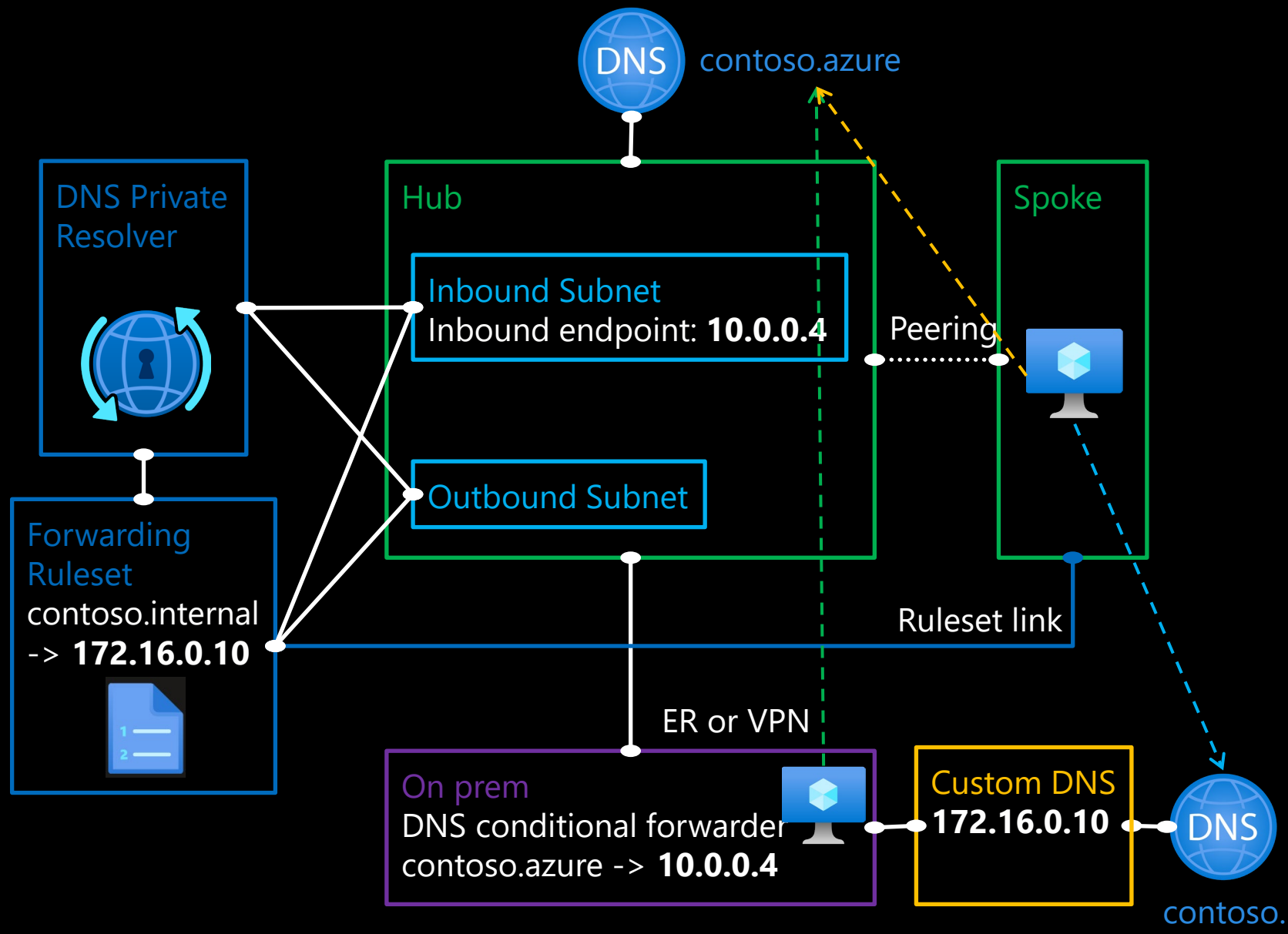
	Price
Azure DNS Private Resolver inbound endpoint	1 endpoint - **$180**/month prorated to hours if discontinued earlier
Azure DNS Private Resolver outbound endpoint	1 endpoint - **$180**/month prorated to hours if discontinued earlier
Azure DNS Private Resolver rulesets	1 ruleset - **$2.50**/month prorated to hours if discontinued earlier

azure.microsoft.com/en-us/pricing/details/dns/

Notes for DNS Private Resolver

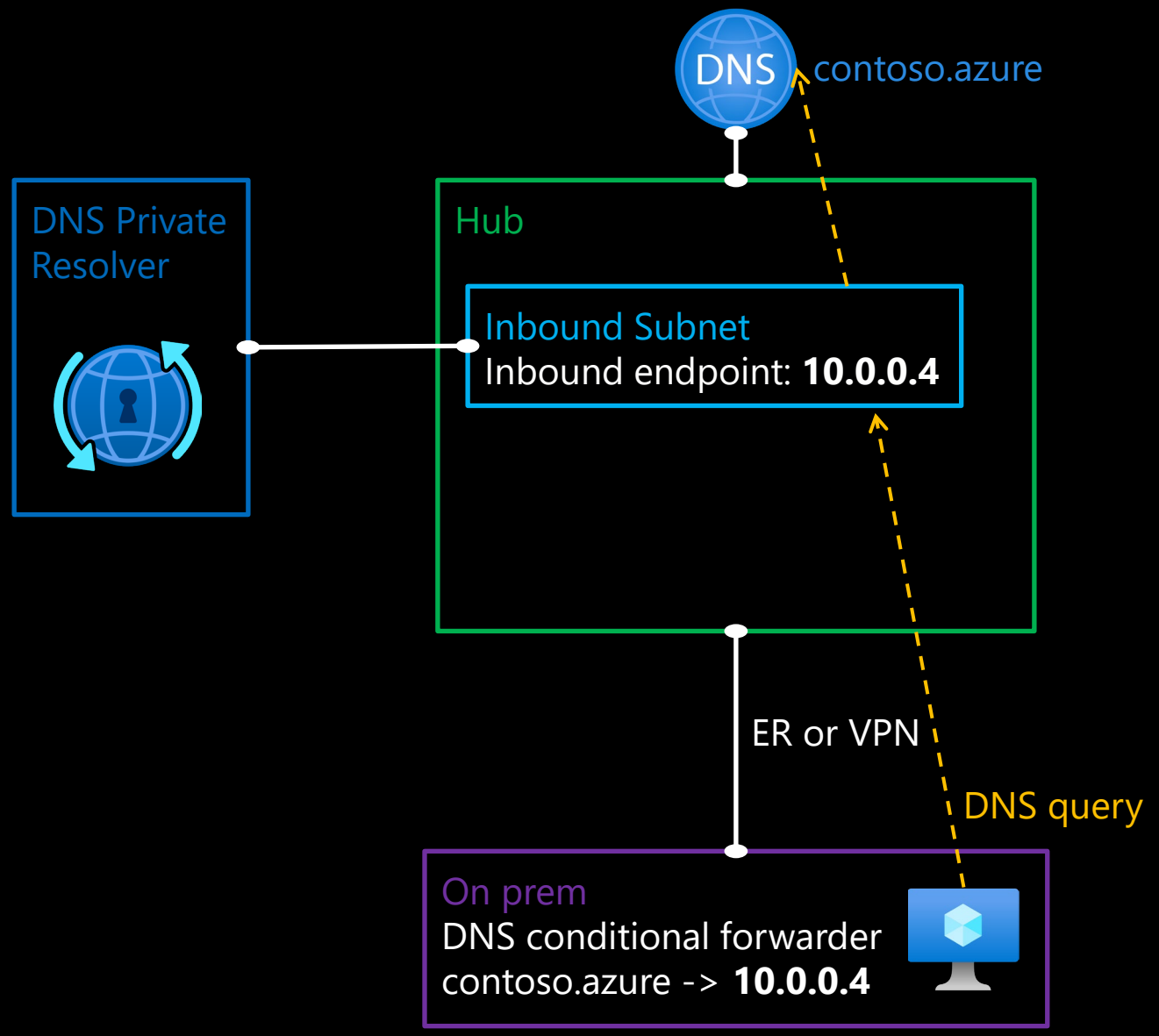
- It can not resolve Private DNS Zones that are not linked to its VNET.
- A DNS resolver can only reference a VNET in the same region.
- VNET can't be shared between multiple DNS resolvers. VNET can only be referenced by a single DNS resolver.

DNS Private Resolver



Resolve domain names:

- From on-prem to Azure
- From Azure to on-prem
- From Azure to Azure



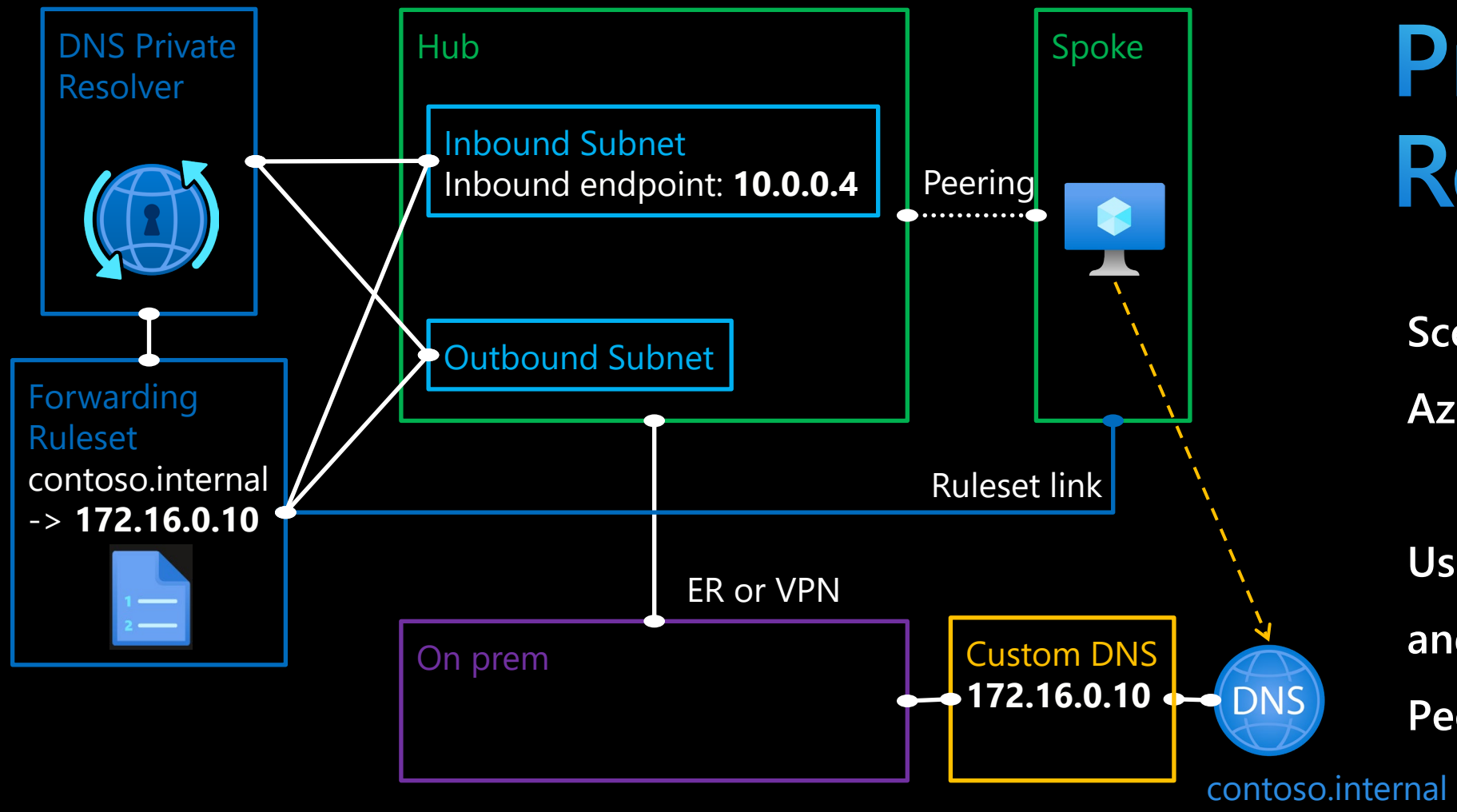
DNS Private Resolver

Scenario 1: On prem to Azure

Inbound endpoint can resolve domain names linked to Hub VNET.

No need for Outbound Endpoint or Ruleset.

DNS Private Resolver



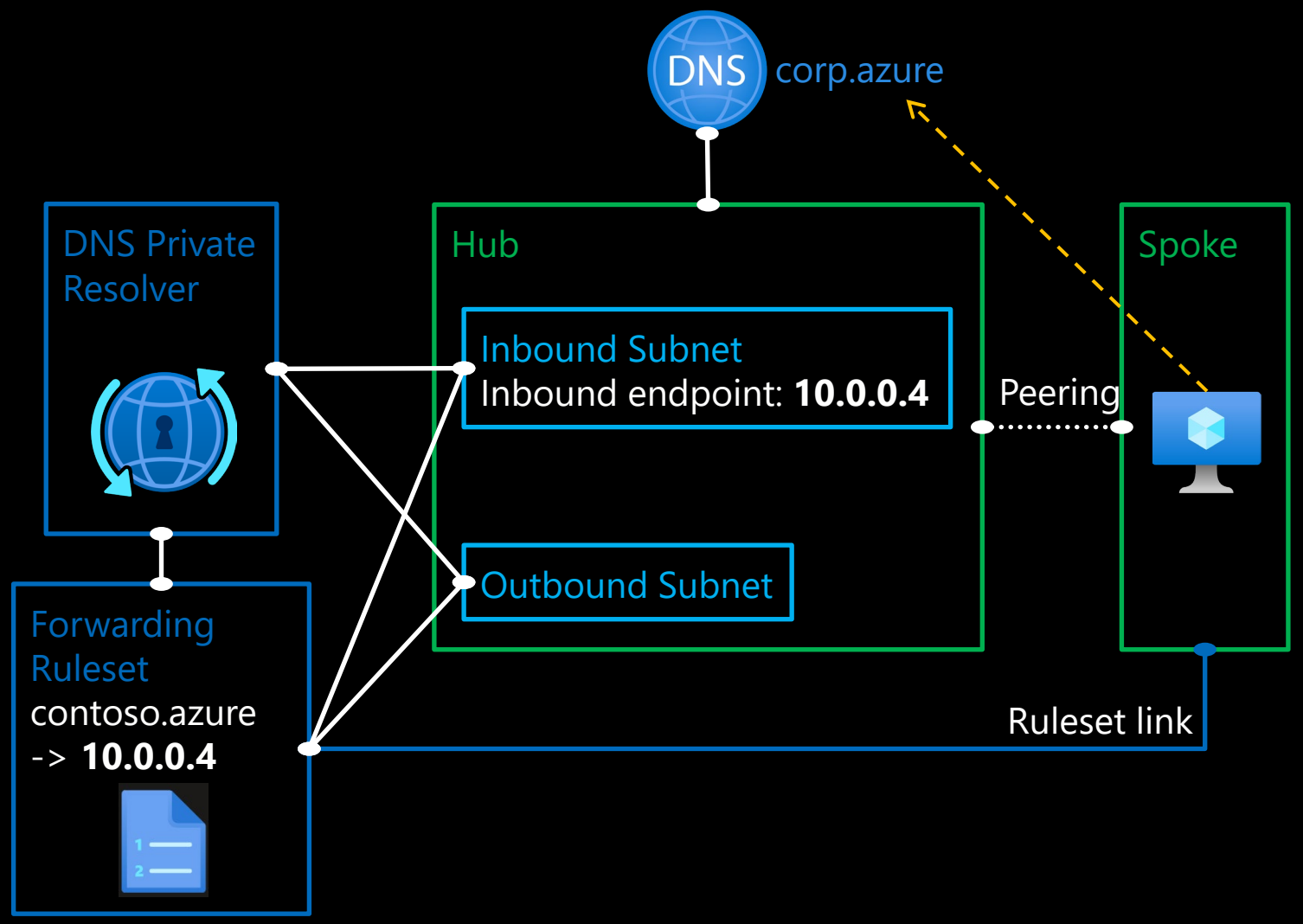
Scenario 2:

Azure to on-prem

Uses Oubound Endpoint and the link.

Peering not required.

DNS Private Resolver



Scenario 3:

Azure to Azure

Uses Oubound Endpoint and the link.

Peering not required.

DNS Private Resolver - Azure to Azure

1) Create Hub VNET with no Subnet
2) Create `Private DNS Zone` called `corp.azure` and an `A record` called `addr1`
3) Link the `Private DNS Zone` to the Hub VNET
4) Create a Linux VM in the Hub VNET called vm-linux-hub
5) Try to resolve the domain name from the vm-linux-hub using `nslookup addr1.corp.azure`, should work
6) Create a Spoke VNET (peering not required) with a Subnet for VMs
7) Deploy a VM in Spoke VNET called vm-linux-spoke
8) Try to resolve the domain name from the vm-linux-spoke using `nslookup addr1.corp.azure`, won't work
9) Create `DNS Private Resolver` (empty)
10) Create `Inbound Endpoint` for DNS Private Resolver in new Subnet
11) Create `Outbound Endpoint` for DNS Private Resolver in new Subnet
12) Create a `Ruleset` (empty)
13) Create a Ruleset rule for `corp.azure`
14) Create Ruleset link to Spoke VNET
15) Try to resolve the domain name from the vm-linux-spoke using `nslookup addr1.corp.azure`, should work